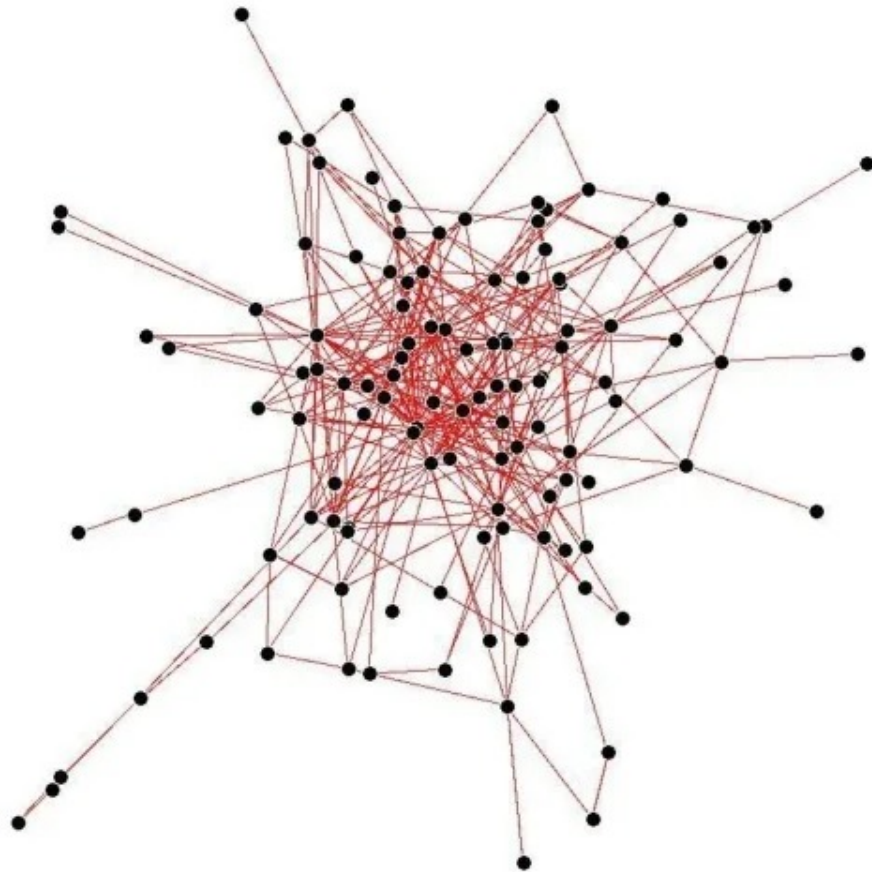
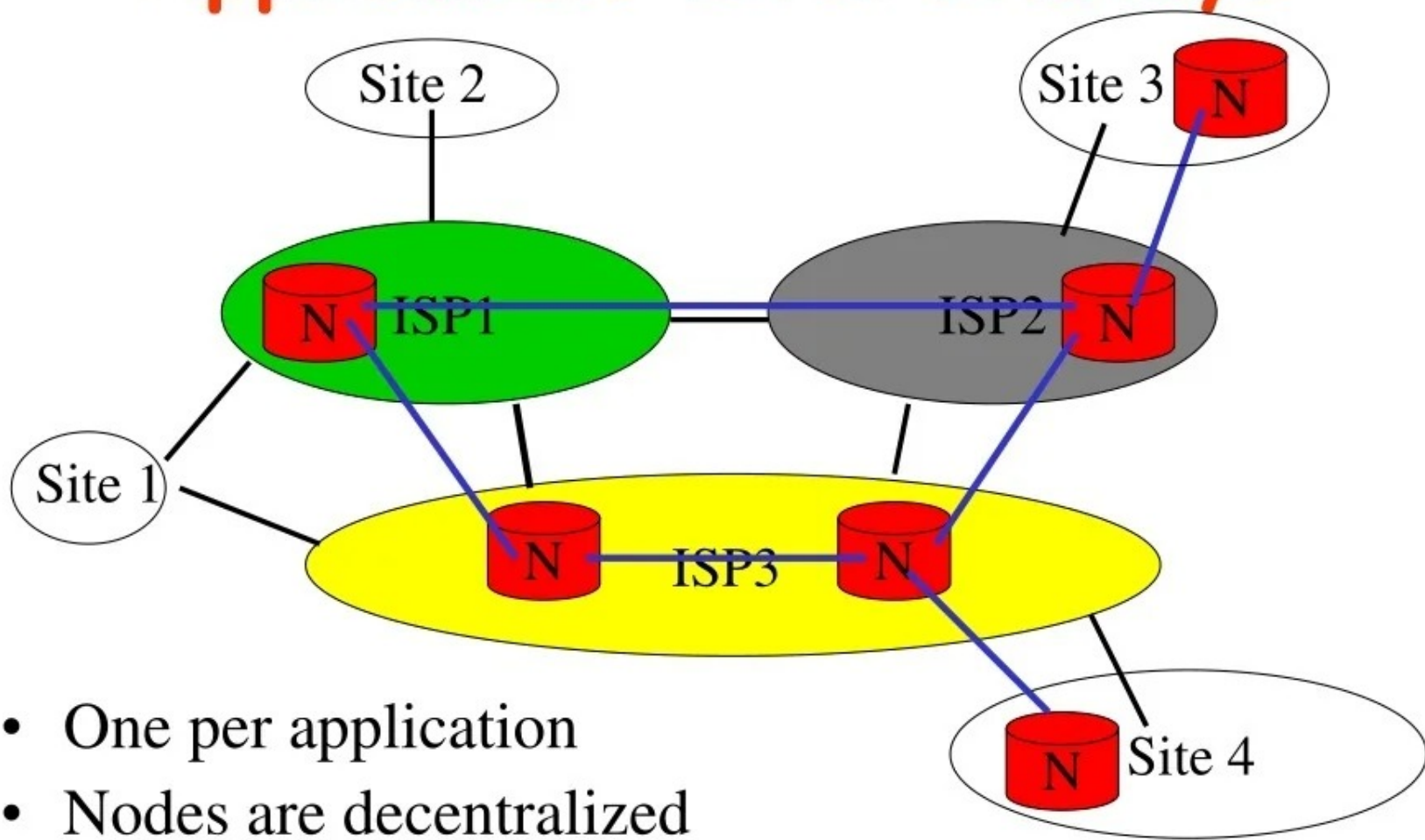


P2P



Application-level overlays



P2P systems are overlay networks without central control

Client/Server Limitations

- Scalability is hard to achieve
- Presents a single point of failure
- Requires administration
- Unused resources at the network edge
- P2P systems try to address these limitations

Client-Server Versus Peer-to-Peer Network Architecture

- A simple distinction
 - Client-server
 - Computers perform asymmetric functions
 - Peers-to-Peer (P2P)
 - Computers perform symmetric functions
- Different architectures offer different benefits
- Pure P2P networks are rare
 - Most P2P networks rely on centralized server for some functions

What is P2P?

...a technology that enables two or more peers to collaborate spontaneously in a network of equal peers by using appropriate information and communication systems without the necessity for central coordination.

- File/information/resource sharing
- Equal peers
- Decentralization

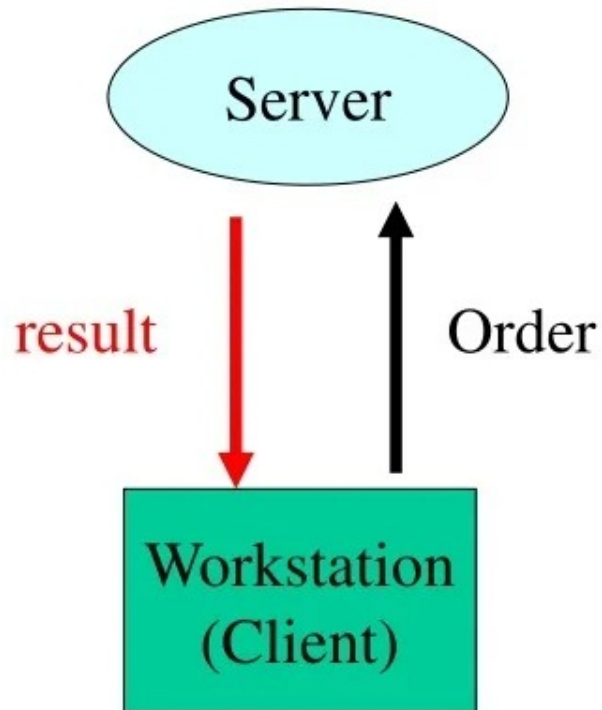


Client/Server Model

Assumption

Workstation is **SO powerless**
that it can not do any task.

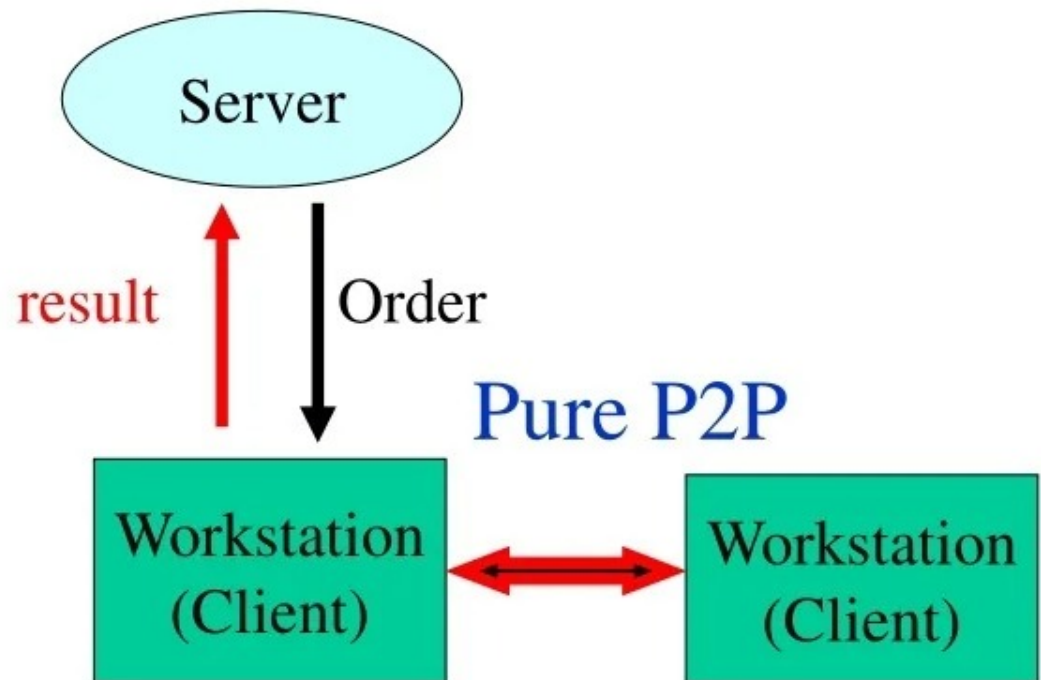
Only **User** (Operator) can
control the workstation



Peer to Peer Model

Assumption

Workstation is **powerful**
enough to do some jobs
Other workstation and server can
remote-control the workstation



P2P Network Features

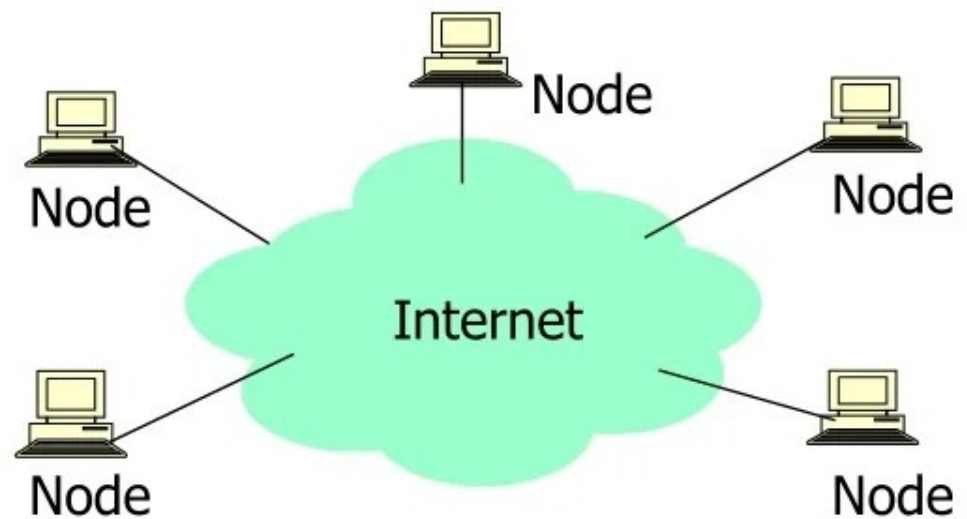
- Clients are also **servers and routers**
 - Nodes contribute content, storage, memory, CPU
- Nodes are **autonomous** (no administrative authority)
- Network is **dynamic**: nodes enter and leave the network “frequently”
- Nodes **collaborate directly** with each other (not through well-known servers)
- Nodes have widely **varying capabilities**

Features of the P2P Computing

- P2P computing is the sharing of computer resources and services by *direct* exchange between systems.
- These resources and services include the exchange of information, processing cycles, cache storage, and disk storage for files.
- P2P computing takes advantage of existing computing power, computer storage and networking connectivity, allowing users to leverage their collective power to the 'benefit' of all.

P2P Architecture

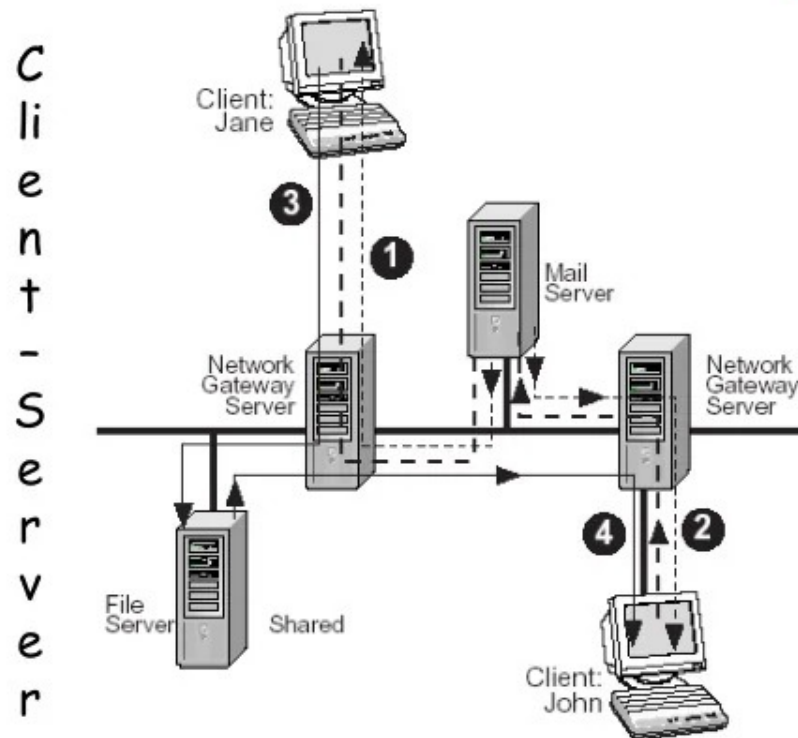
- All nodes are both clients and servers
 - Provide and consume data
 - Any node can initiate a connection
- No centralized data source
 - “The ultimate form of democracy on the Internet”
 - “The ultimate threat to copy-right protection on the Internet”



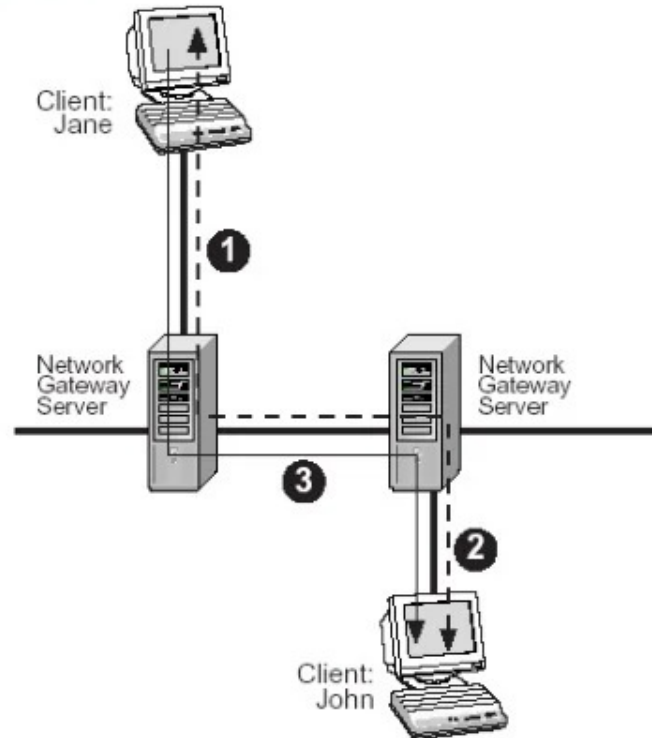
P2P Network Characteristics

- Clients are also **servers and routers**
 - Nodes contribute content, storage, memory, CPU
- Nodes are **autonomous** (no administrative authority)
- Network is **dynamic**: nodes enter and leave the network “frequently”
- Nodes **collaborate directly** with each other (not through well-known servers)
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Client-Server vs. Peer-to-Peer Example

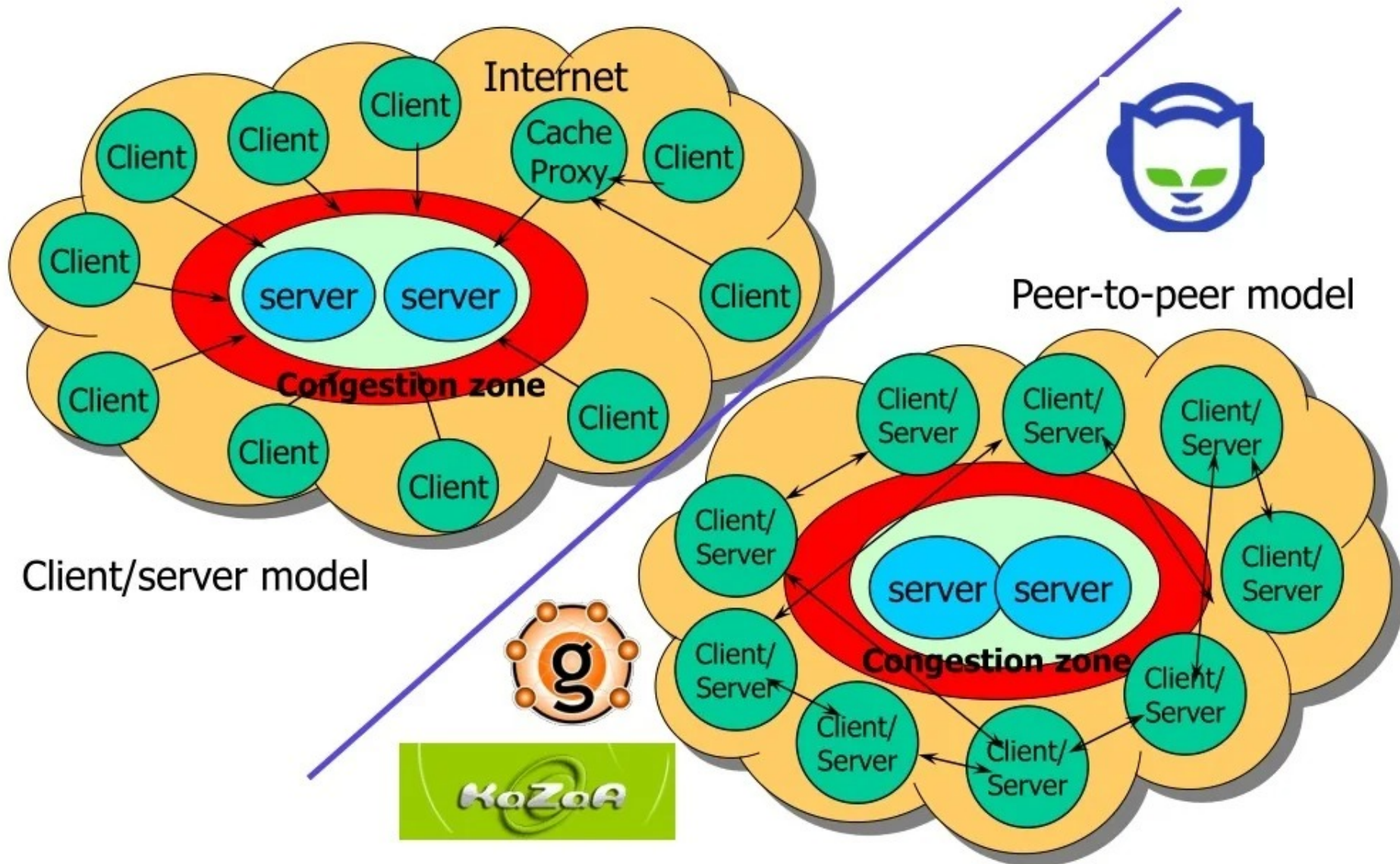


- ① John mails Jane: "I need the file "Plan.doc." Jane reads when synchronizing her inbox.
- ② John "pulls" Jane's mail reply: "I will place Plan.doc on group's shared drive."
- ③ Jane uploads file to shared drive.
- ④ John downloads the file.



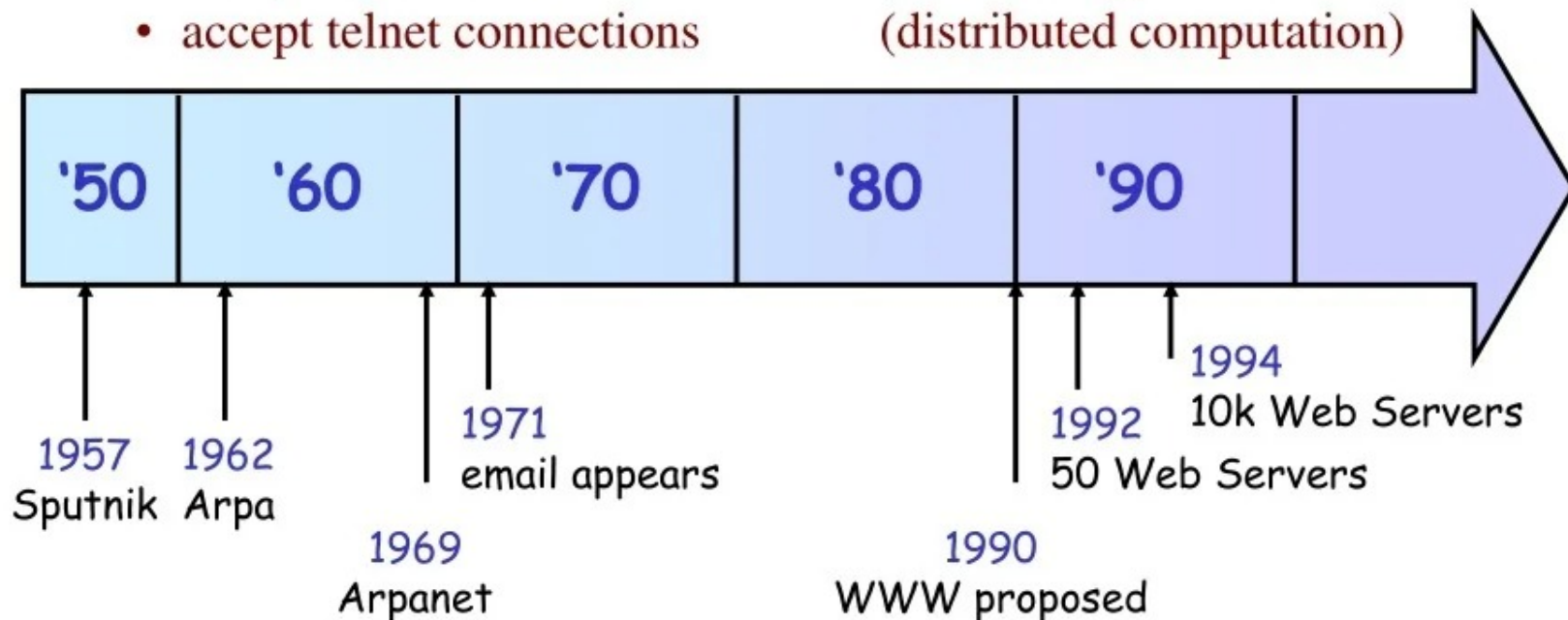
- ① Instant Message (IM), John to Jane: "I will need the file Plan.doc."
- ② IM, Jane to John: "It's on my shared storage now."
- ③ John downloads the file from Jane's computer.

Large-Scale Data Sharing: P2P



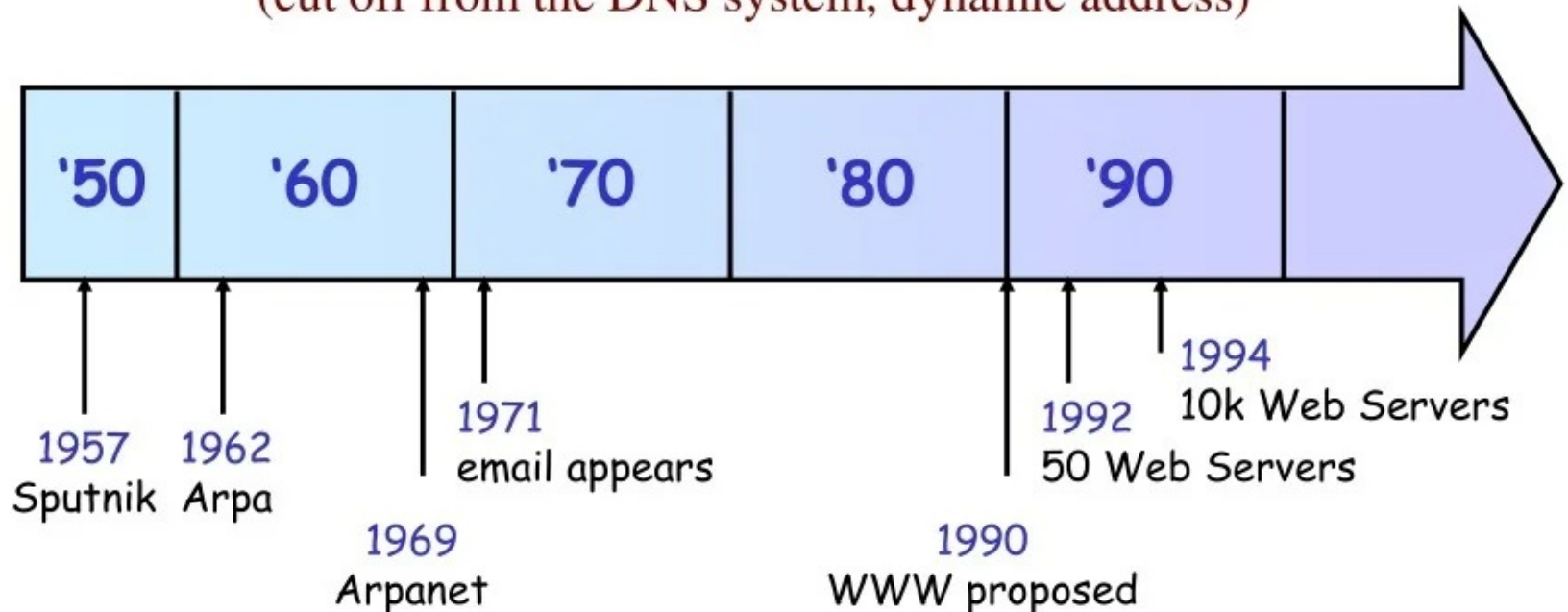
P2P History: 1969 - 1995

- 1969 – 1995: the origins
 - In the beginning, all nodes in Arpanet/Internet were peers
 - Every node was capable to
 - perform routing (locate machines)
 - accept ftp connections (file sharing)
 - accept telnet connections (distributed computation)



P2P History: 1995 - 1999

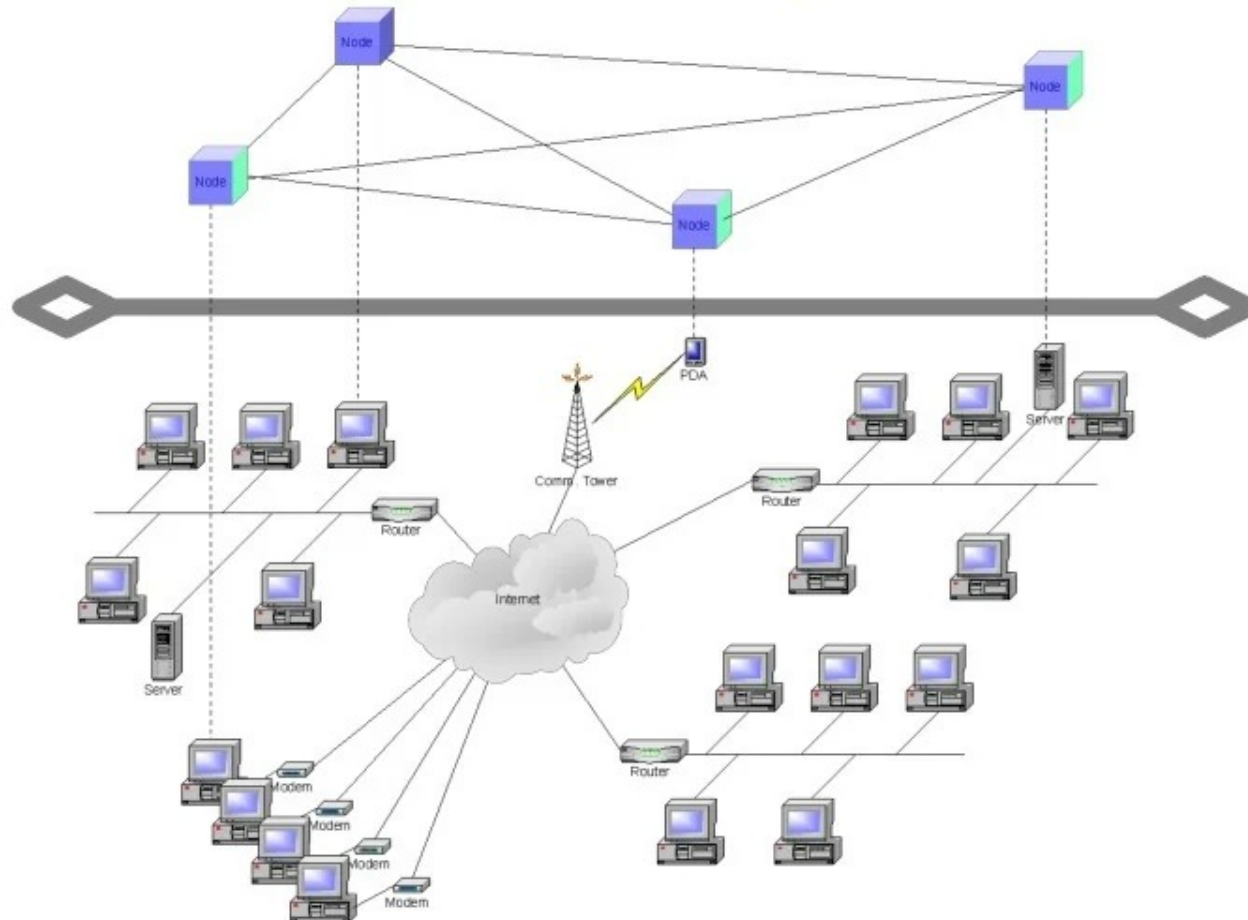
- 1995 – 1999: the Internet explosion
 - The original “state of grace” was lost
 - Current Internet is organized hierarchically (client/server)
 - Relatively few servers provide services
 - Client machines are second-class Internet citizens (cut off from the DNS system, dynamic address)



P2P History: 1999 - today

- 1999 – today: the advent of Napster
 - Jan 1999: the first version of *Napster* is released by Shawn Fanning, student at Northeastern University
 - Jul 1999: Napster, Inc. founded
- In short time, Napster gains an enormous success, enabling millions of end-users to establish a file-sharing network for the exchange of music files
 - Jan 2000: Napster unique users > 1.000.000
 - Nov 2000: Napster unique users > 23.000.000
 - Feb 2001: Napster unique users > 50.000.000

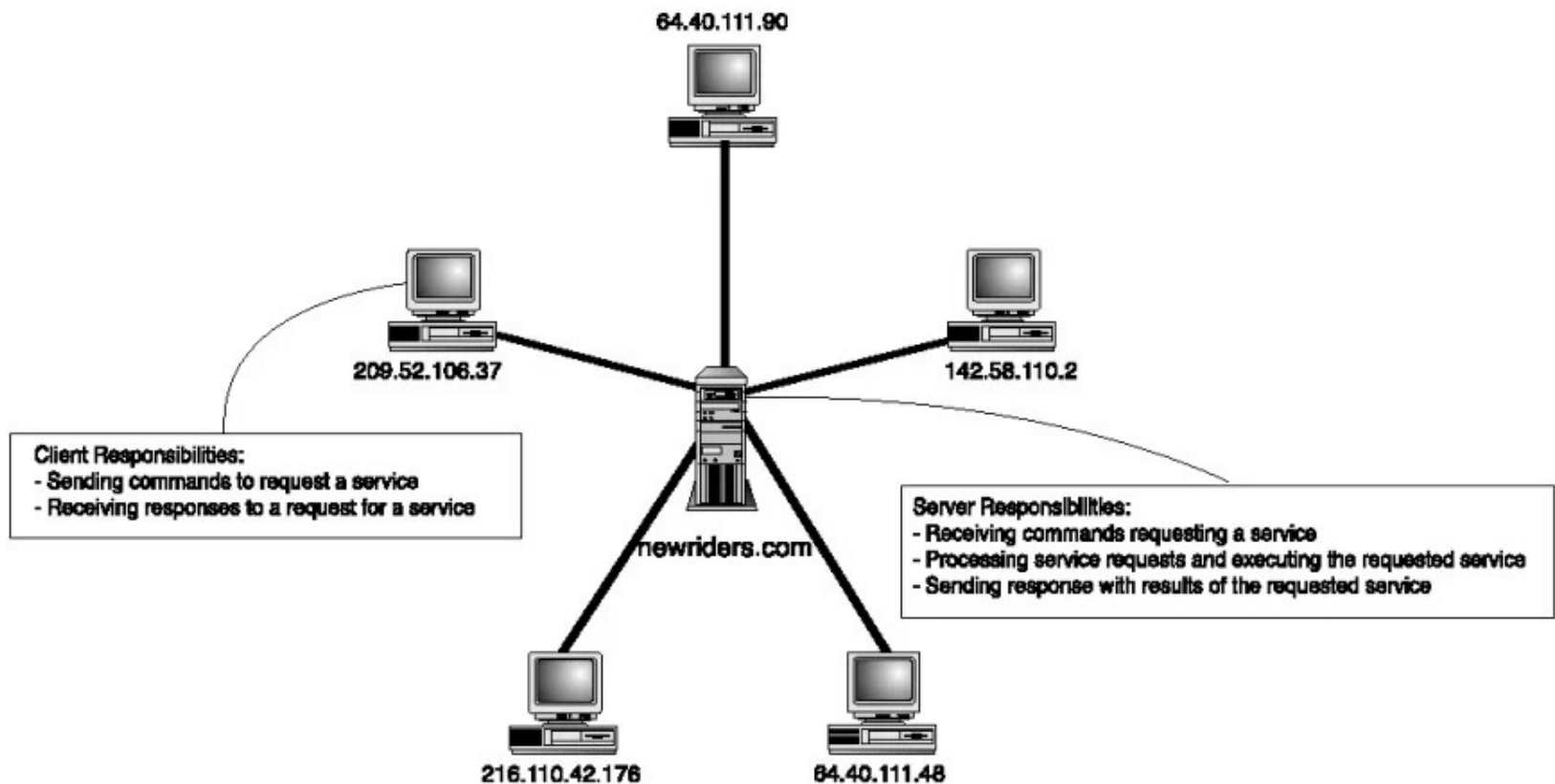
p2p \pir too pir\ n. a virtual network of functionally similar nodes created using an alternate, often private, namespace



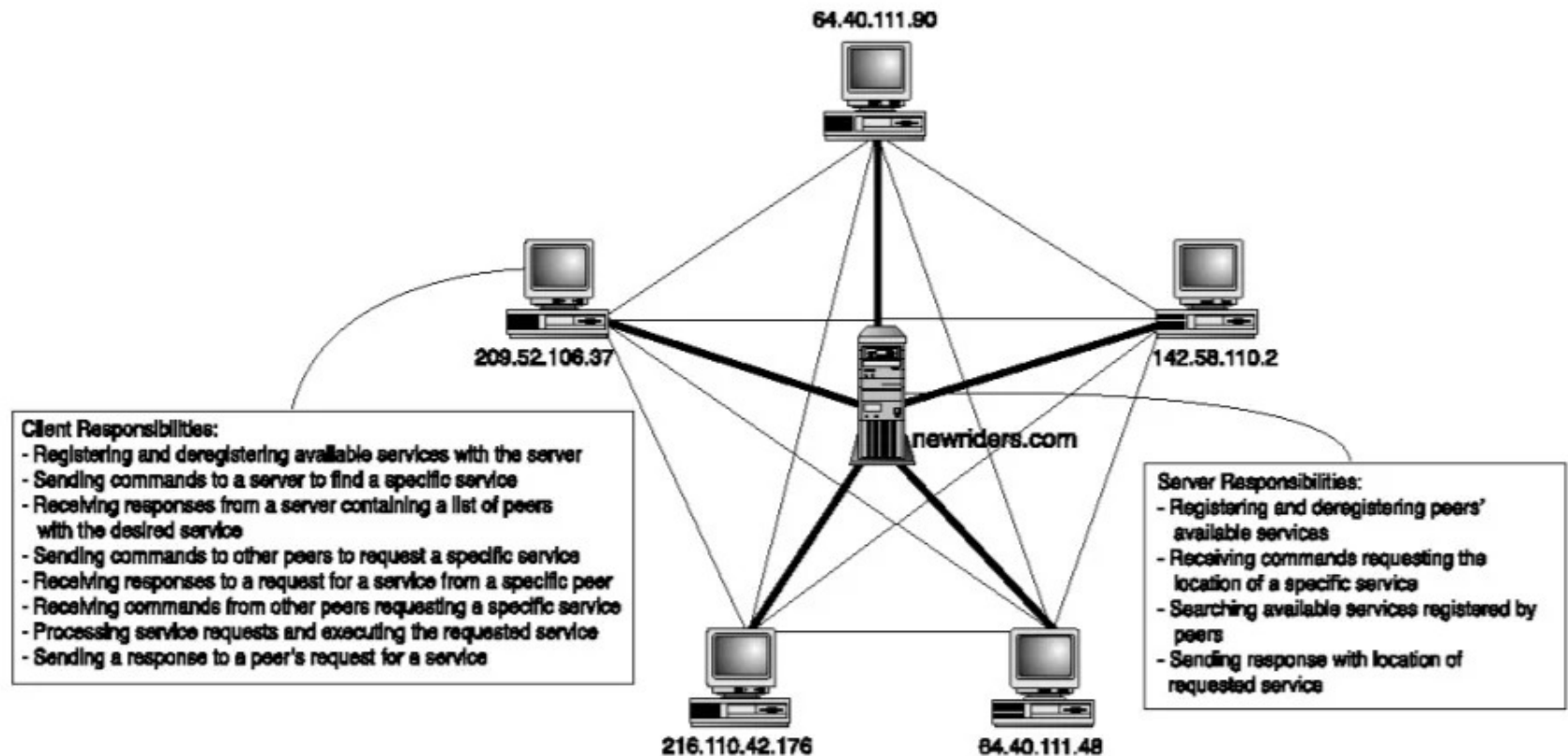
P2P Benefits

- Efficient use of resources
 - Unused bandwidth, storage, processing power at the edge of the network
- Scalability
 - Since every peer is alike, it is possible to add more peers to the system and scale to larger networks
 - Consumers of resources also donate resources
 - Aggregate resources grow naturally with utilization
- Reliability
 - Replicas
 - Geographic distribution
 - No single point of failure
 - E.g., the Internet and the Web do not have a central point of failure.
 - Most internet and web services use the client-server model (e.g. HTTP), so a specific service does have a central point of failure
- Ease of administration
 - Nodes self organize
 - No need to deploy servers to satisfy demand – confer (compare, c.f.) scalability
 - Built-in fault tolerance, replication, and load balancing

The traditional network architecture



A hybrid P2P network architecture



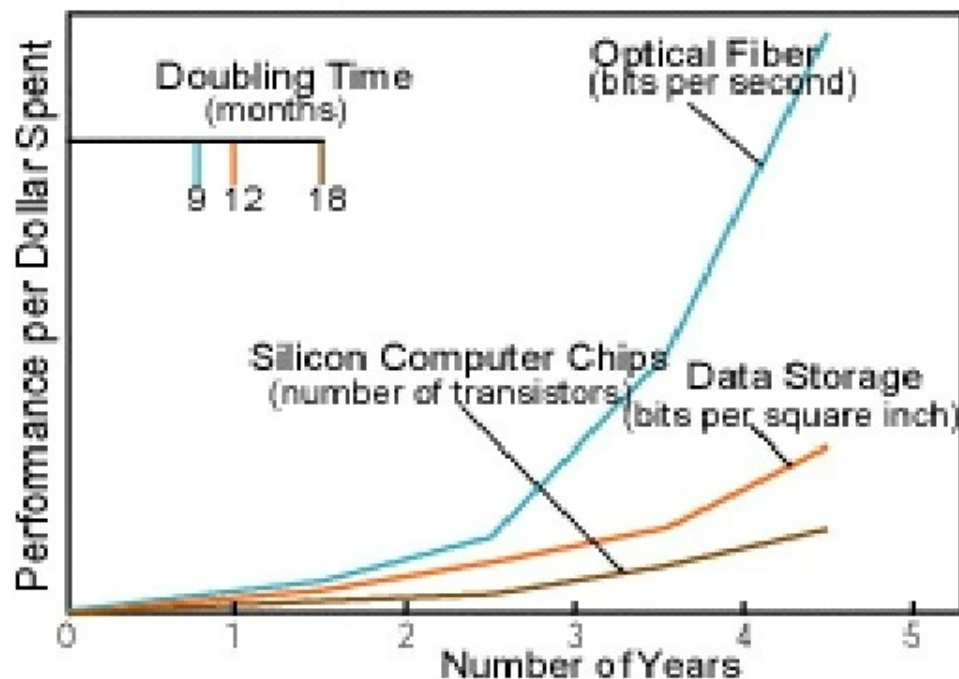
P2P must be disruptive...

- Peer-to-peer (p2p): third generation of the Internet
- 1st generation: “raw” Internet
- 2nd generation: the Web
- 3rd generation: making new services to users cheaply and quickly by making use of their PCs as active participants in computing processes
- P2P doing this in “disruptive” ways

Bandwidth and Storage Growth

> Moore's Law

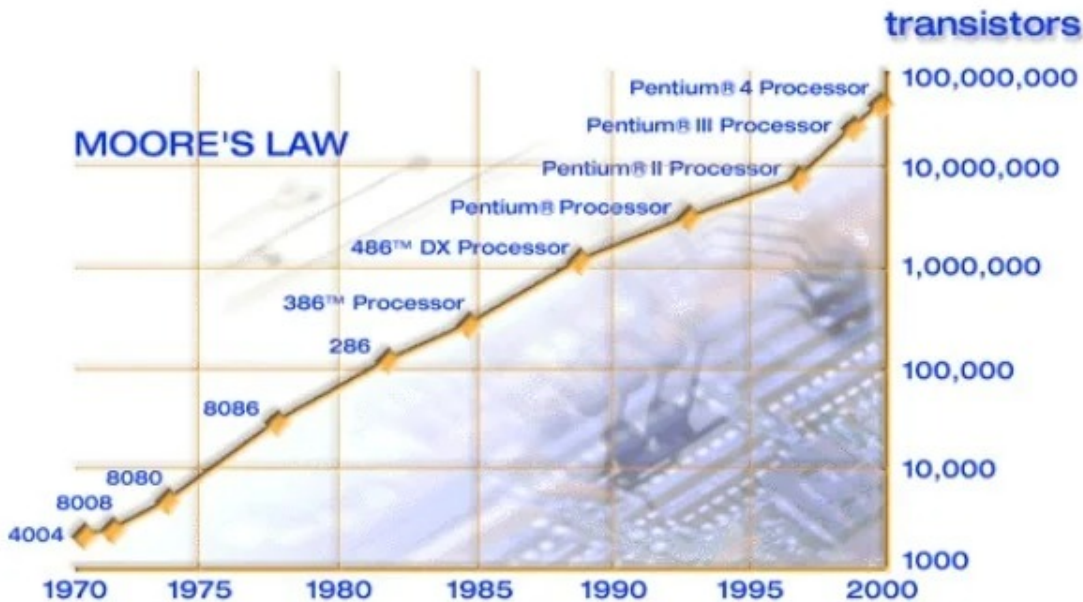
- Network, Storage and Computers
 - Network speed doubles every 9 months
 - Storage size doubles every 12 months
 - Computer speed doubles every 18 months
- 1986 to 2000
 - Computers : X 500
 - Storage : X 16,000
 - Networks : X 340,000
- 2001 to 2010
 - Computers : X 60
 - Storage : X 500
 - Networks : X 4000



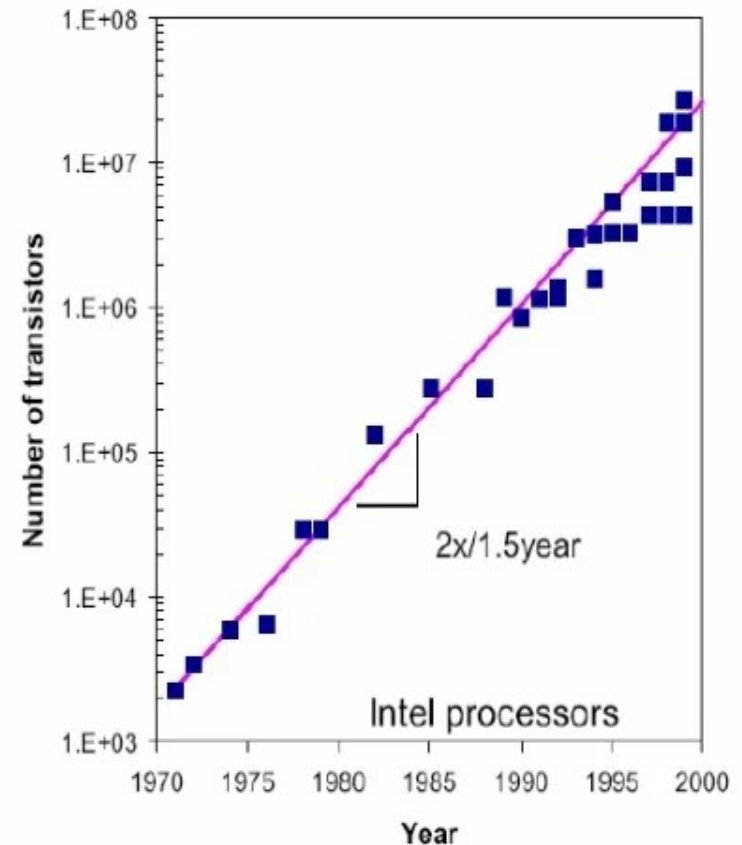
Graph from *Scientific American* (Jan 2001) by Cleo Villett, source Vined Khoslan, Kleiner, Caufield and Perkins.

Moore's Law

- In 1965, Gordon Moore predicted that the number of transistors that can be integrated on a die would double every 18 to 14 months
 - i.e., grow exponentially with time
- Amazing visionary – million transistor/chip barrier was crossed in the 1980's.
 - 2300 transistors, 1 MHz clock (Intel 4004) - 1971
 - 42 Million, 2 GHz clock (Intel P4) - 2001
 - 140 Million transistor (HP PA-8500)

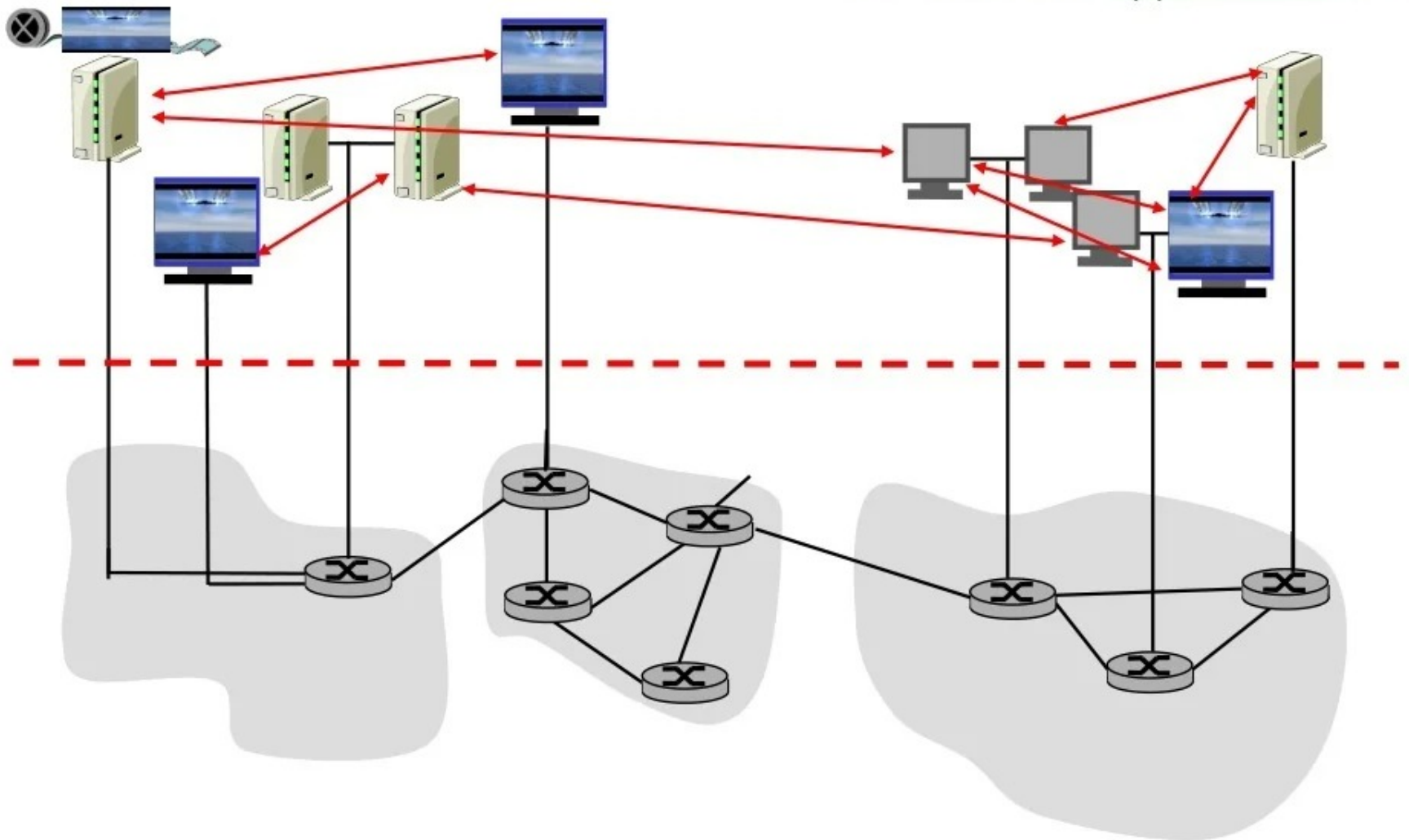


Source: Intel web page (www.intel.com)



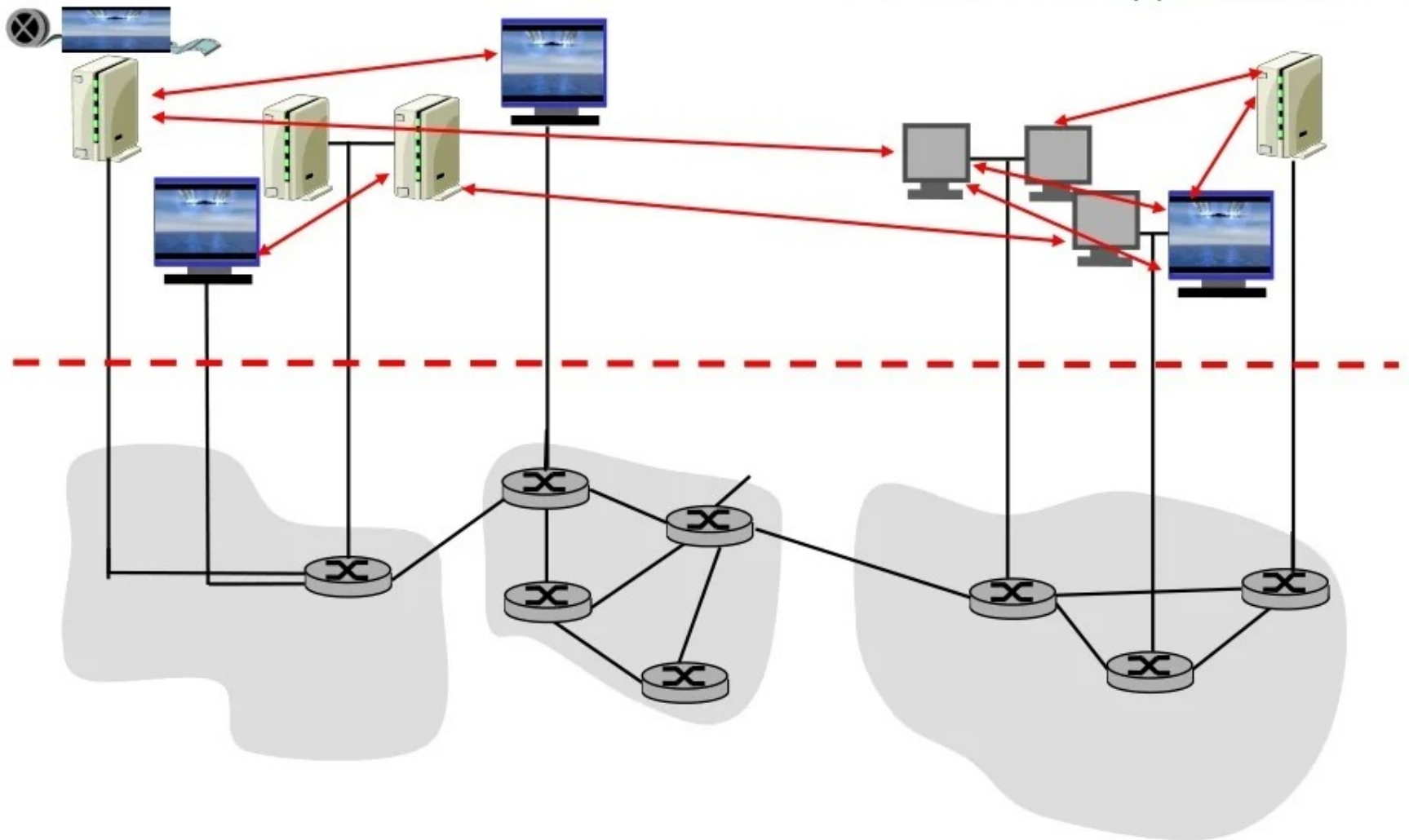
Peer-peer networking

Focus at the application level

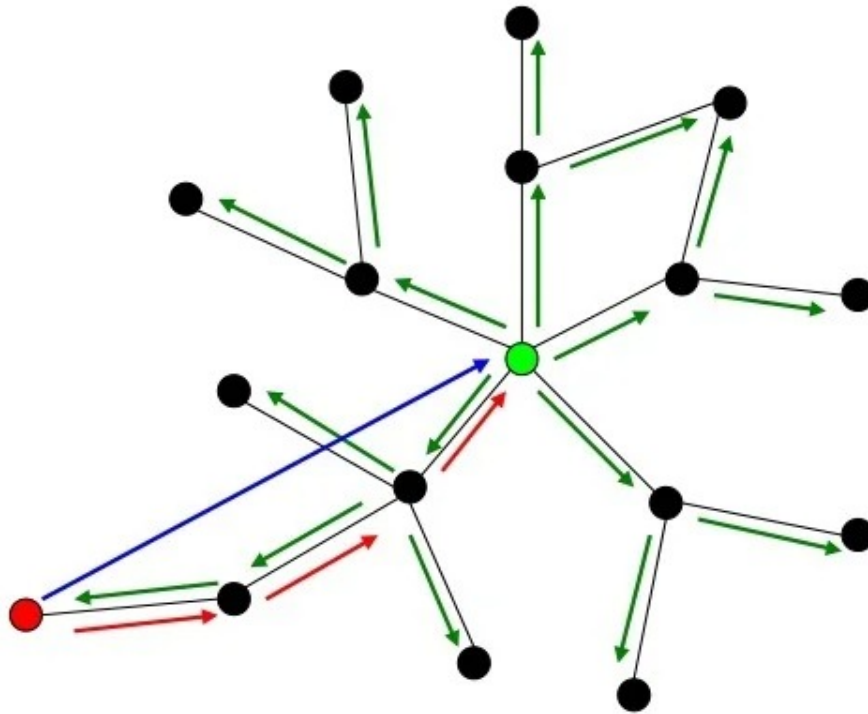


Peer-peer networking

Focus at the application level



How it works



Classification of the P2P Systems

Three main categories of systems

- **Centralized systems:** peer connects to server which coordinates and manages communication. e.g. SETI@home
- **Brokered systems:** peers connect to a server to discover other peers, but then manage the communication themselves (e.g. Napster). This is also called *Brokered P2P*.
- **Decentralized systems:** peers run independently with no central services. Discovery is decentralized and communication takes place between the peers. e.g. Gnutella, Freenet



True P2P

What P2P is good for?

- Community Web network
 - Any group with specific common interests, including a family or hobbyists, can use lists and a Web site to create their own intranet.
- Search engines
 - Fresh, up-to-date information can be found by searching directly across the space where the desired item is likely to reside
- Collaborative development
 - The scope can range from developing software products to composing a document to applications like rendering graphics.

P2P Application Areas

- **Communication**
 - AOL Instant Messenger
 - ICQ
- **Remote Collaboration** (Shared File Editing, Audio-video Conferencing)
 - Jabber
 - Shared whiteboard
- **Multiplayer Games**
 - Unreal Tournament, DOOM
- **Streaming** (Application-level Multicast)
 - Narada
 - Yoid
 - NICE, CAN-Multicast, Scribe
- **Distributed Computing**
 - SETI@home
- **File Sharing**
 - Napster
 - Gnutella, Freenet, LimeWire
 - KazaA, Morpheus
- **Ad-hoc networks**

File-sharing vs. Streaming

- File-sharing
 - Download the *entire* file first, then use it
 - Small files (few Mbytes) □ short download time
 - A file is stored by one peer □ one connection
 - No timing constraints
- Streaming
 - Consume (playback) as you download
 - Large files (few Gbytes) □ long download time
 - A file is stored by multiple peers □ several connections
 - Timing is crucial

Example P2P Applications

- The following areas are detailed in the following slides:
 - Instant Messaging
 - File exchange
 - Collaboration
 - MIPS sharing
 - Lookup services
 - Mobile ad hoc communication
 - Content Distribution
 - Middleware

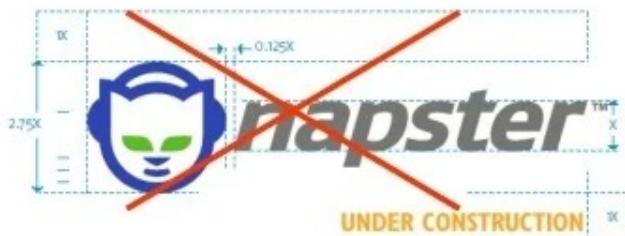
Instant Messaging

- A convenient way of communicating with a small group of selected people (e.g., friends, family members, etc.)
- Usually, a central server is used to store user profiles and to have a list of registered users
- While communication takes place between the peers, searching for other people is done using the server
- One of the reasons why a server is needed is the ability to send messages to other persons (i.e., peers)
- If the target peer is not online, the system has to store the message until the target peers becomes online again
- This would be, of course, also possible with a server-less P2P system, but the price would be an increased complexity and a certain probability of messages getting lost
- Examples for such systems are:
 - Napster
 - ICQ
 - threedegrees
 - Jabber

File exchange



- There is little dispute about the usefulness of P2P file sharing applications
- While downloading files is always done directly between peers (or via a proxy peer to enable anonymity), the way of searching for these files differs in many P2P applications
- Some use central servers (e.g., Napster) while others send search requests
- directly to other peers (e.g., Gnutella, Freenet, & FastTrack)



Collaboration

- This is not a typical example for the usefulness of P2P technology
- It is about having people having the same view or different views on shared information
- This would typically call for a server storing this information
- This way, the information is available to all members without the necessity of having the information provider or contributor online or the data distributed to all other participants
- But there are use cases where P2P technology comes in mobile phones
- One example could be ad hoc collaboration of devices in an environment where no connection to a server exist
 - E.g., people are meeting in a place where no connection to the Internet is available
- In this scenario, people would communicate (and collaborate) in a server-less P2P manner
- One perfect example for this use case is Groove
 - It uses a server to store shared information but is also able to provide collaboration services without the existence of such a server

MIPS sharing

- One of the major assets of the Internet is its combined processing power
 - which is currently vastly under-utilized
- To utilize these resources, user are asked to download and install programs that are able to do a small part of a complex computation while the computer is not used
 - E.g., while the screen saver is running
- Examples for MIPS sharing systems are:
 - Seti@HOME
 - Genome@HOME
- In this category of P2P applications, the social aspect is very important
- Were it not for the search for extraterrestrial life or cancer research, not many people would be willing to share their processing power
 - Hence, there must an incentive for users to share computer resources, be it money, public well-fare or the like
- Furthermore, this type of P2P application can only function with a central server that is coordinating the distribution of computation tasks and the validation of the results

Lookup services

- Most of the scientific P2P research is done in the area of lookup services
- This is not very surprising because searching is one of the major challenges in P2P networks
- Most of the P2P systems that are optimized for lookup services are using distributed hash tables (DHT)
 - which are capable of searching with logarithmic complexity
- The drawback of most of these systems is the fact that they are only able to search for numbers
 - In case they are searching for strings, they are searching for numerical representations of these strings
- Examples for such systems are:
 - PAST
 - Chord
 - P-Grid

Mobile ad hoc communication

- Ad hoc communication, especially when it is done among mobile devices
 - I.e., the devices are connected directly via a wireless communication link
 - This is the best example for the usefulness of the P2P paradigm
- Devices connect to each other in an ad hoc manner
- Due to the limited communication capabilities of mobile devices (such as mobile phones or handheld devices), frequent disconnections may occur
- When mobile devices are connected together, there is no guarantee that a central server may be available
- Hence, ad hoc mobile communication must not rely on the existence of such a server
- All these characteristics also apply to the P2P paradigm
- There exists only a small number of P2P systems that can be used in conjunction with small devices:
 - GnuNet
 - JXME (JXTA for J2ME - the Java 2 Mobile Environment)

Content Distribution

- P2P can also be used for the distribution of information or files
 - sometimes called ESD - electronic software distribution
- Instead of having a central source that emits files to the destination computers directly, a P2P network may disseminate files while avoiding hot spots in the network
- The load (i.e., the bandwidth, CPU power, throughput, etc.) is distributed over the whole network
- This concept is successfully used, for example, by Intel where software is distributed to international branches in a P2P style
- This system can be compared to a push system
- The advantage is that there is no need for a fixed environment of push server and proxies

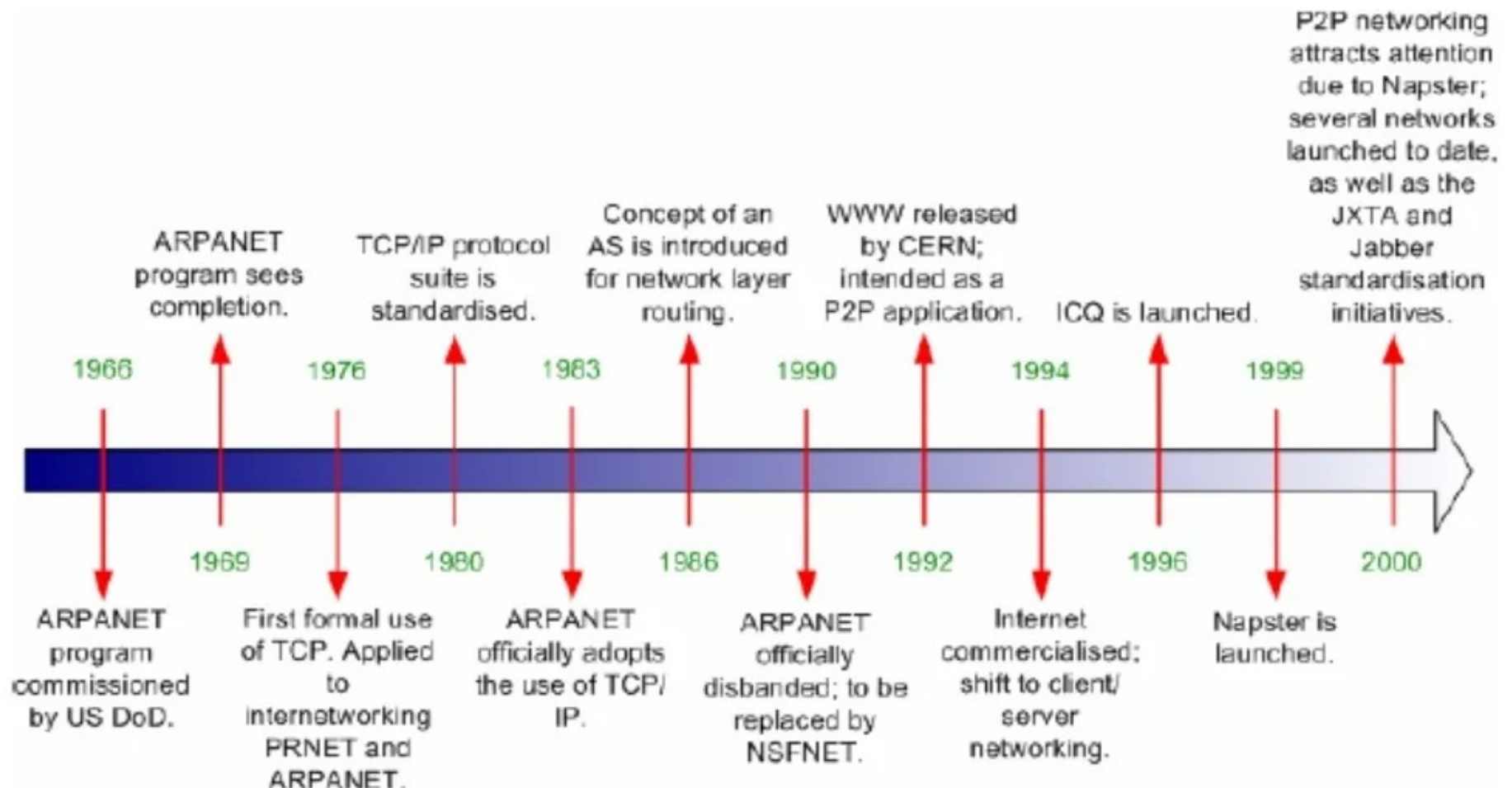
Middleware

- The most demanding use case for a P2P system is its use as a middleware platform. P2P middleware systems provide services such as distributed search or peer discovery to higher-level applications
- Different kind of P2P systems have a limited set of application domains
- Depending on the structure (or topology) of the P2P network, various use cases may become feasible or impossible
- If a P2P system is needed as a middleware, the use case turns the balance which P2P system best fits the requirements
- Only a few P2P systems may be used as a P2P middleware platform:
 - JXTA
 - Omnix

Port Numbers Used by Various P2P Applications

P2P application	Port
Bittorrent	default 6881-6889 TCP/UDP
edonkey	fully configurable default 4662/TCP, 5737/UDP
Gnutella	default 6346/6347 TCP/UDP
Morpheus	default 6346/6347 TCP/UDP
Kazaa	default 1214 TCP/UDP
EMule	fully configurable default 4662/TCP 4672/UDP
WinMx	fully configurable default 6699/TCP 6257/UDP
Limewire	default 6346/6347 TCP/UDP
Napster	client default 6699 TCP alternate 6600-6699 TCP
BearShare	default 6346 TCP/UDP

Timeline of P2P networking evolution



P2P Technical Challenges

- Peer identification
- Routing protocols
- Network topologies
- Peer discovery
- Communication/coordination protocols
- Quality of service
- Security
- Fine-grained resource management